

Chapter 8: Advanced Theories of Covalent Bonding

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CHM111
Lecture 8



Part I

Why We Need Advanced Bonding Theories

Outline of Part I: Why We Need Advanced Bonding Theories

Limitations of Lewis Structures and VSEPR

Outline of Part I: Why We Need Advanced Bonding Theories

Limitations of Lewis Structures and VSEPR

Modeling Orbitals and Electron Distribution

What Lewis Structures Cannot Explain

Review: What We Know So Far

- ▶ Lewis structures show **how many** bonds exist and where lone pairs sit.
- ▶ VSEPR theory predicts **molecular geometry** from electron-pair repulsions.

What These Models Fail to Explain

- ▶ **Why** bond angles are 109.5° , 120° , or 180° — VSEPR predicts angles empirically, but doesn't describe the *orbitals* producing them
- ▶ Why O_2 is attracted to a magnet, even though its Lewis structure shows no unpaired electrons
- ▶ The **electronic structure** of molecules with delocalized electrons

Roadmap: from models of *electron pairs* (Lewis, VSEPR) to models of *orbitals* (VB, hybridization) and *electron distributions* (MO theory).

Four Bonding Models: What Each Is Useful For

Each model is a tool, optimized for a different question

Model	Useful for explaining	Cannot explain
Lewis	connectivity, electron counting	geometry, magnetism, orbitals
VSEPR	molecular geometry	electron distribution, orbitals
VB	bonding via orbital overlap	full delocalization
MO	delocalization, magnetism	geometry less directly

(VB: Valence Bond), (MO: Molecular Orbital) — Both are introduced in this lecture.

 Notice that scientists do not simply abandon models that fail in one aspect.

We instead **add new ones** to address their deficiencies, keeping the old models as long as they continue to answer questions and aid understanding.

Chemists choose whichever model fits the question at hand.

Electron Magnetism: Diamagnetism v. Paramagnetism

Origin of Electron Magnetism

Electron magnetic moments arise from two quantum factors:

- ▶ **Electron spin** (m_s): an intrinsic quantum property
- ▶ **Orbital angular momentum** (L): associated with an electron's motion in an atomic orbital

Both contributions produce an electron magnetic dipole moment.

Paramagnetic Substances:

have unpaired electrons
↓
electron magnetic moments do not cancel
↓
permanent *atomic* magnetic moment
↓
atom aligns with an external field
↓
weak *attraction*

Diamagnetic Substances

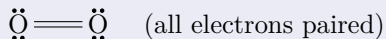
have all electrons paired
↓
electron magnetic moments cancel
↓
no permanent *atomic* magnetic moment
↓
external field induces an opposing moment
↓
weak *repulsion*

The O₂ Paradox: A Magnetic Mystery

Predict First

Based only on the Lewis structure of O₂, would you expect O₂ to be attracted to a magnet?

The Lewis Structure of O₂



Lewis predicts **all paired** $\Rightarrow$ should be **diamagnetic**.

The Experimental Reality

Liquid O₂ is **visibly attracted** to a magnet — it is **paramagnetic**. This requires **unpaired electrons**. (N₂, also double-bonded, is diamagnetic.)

Only **MO theory** can fully explain this observation.

Part II

Valence Bond Theory

Outline of Part II: Valence Bond Theory

Valence Bond Theory

Outline of Part II: Valence Bond Theory

Valence Bond Theory

What Is Valence Bond Theory?

From Lewis to VB

Lewis asks: *which* atoms are connected?

VB asks: *how* do atomic orbitals overlap to create the connection?

Core Idea of Valence Bond (VB) Theory

A **covalent bond forms when occupied atomic orbitals on two atoms overlap**, allowing electrons with opposite spins to pair in the overlap region.

Each pair of overlapping orbitals forms one covalent bond (in the form of a σ or π bond).

Example: H₂ Formation

- ▶ Each H atom has one electron in a 1s orbital; as two H atoms approach, the 1s orbitals **overlap**.
- ▶ The bond is a **region of increased probability density** between the nuclei — not a stationary pair of electrons in a fixed spot.

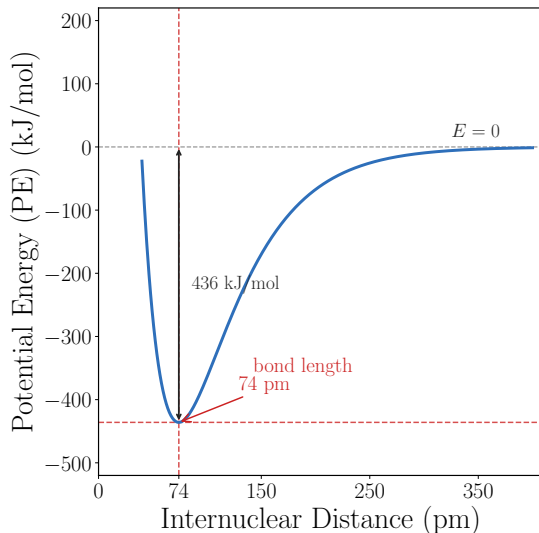
Greater orbital overlap $\Rightarrow$ stronger bond.

Bond Formation: Potential Energy (PE) vs. Internuclear Distance

Potential Energy (PE) Curve for H_2

- ▶ At large separation: atoms are independent; $PE = 0$.
- ▶ As atoms approach: attractive forces lower PE.
- ▶ At **equilibrium bond length** (74 pm): PE is at a minimum (and system is $\therefore$ most stable).
- ▶ Closer than 74 pm: nuclear repulsion raises PE sharply.

The **bond dissociation energy** (436 kJ/mol for H_2) equals the depth of this energy well.



Determining the Type of Covalent Bond

Not All Orbital Overlap Is the Same

Key Consequence of Valence Bond Theory

- ▶ A covalent bond forms when occupied atomic orbitals overlap, allowing electrons with opposite spins to pair in the overlap region.
- ▶ **Each pair of overlapping orbitals forms one covalent bond.**
- ▶ The *geometry* of that overlap determines the type of bond that forms.

Different geometry of orbital overlap $\implies$ Different covalent bond

head-on overlap $\rightarrow$ “ σ bond”

side-by-side overlap $\rightarrow$ “ π bond”

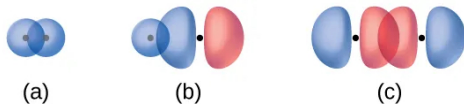
Sigma Bonds (σ)

Definition: Sigma Bond (σ bond)

A **sigma bond** is a covalent bond in which electron density is concentrated **along the internuclear axis** (end-to-end overlap).

How σ Bonds Form

- ▶ $s + s$ overlap (e.g., H–H)
- ▶ $s + p$ overlap (e.g., H–F)
- ▶ $p + p$ end-to-end overlap
- ▶ Hybrid + hybrid overlap (most C–C, C–N, etc.)



Sigma (σ) bonds can form from the overlap of any of the following sets of orbitals:

- (a) two s orbitals
- (b) an s orbital and a p orbital
- (c) two p orbitals

The dots indicate the locations of the **atomic nuclei**.

Pi Bonds (π)

Definition: Pi Bond (π bond)

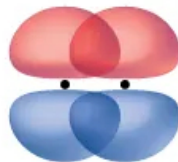
A **pi bond** forms by **side-by-side** (lateral) overlap of two *p* orbitals. Electron density lies *above and below* the internuclear axis, with a **node** along the axis itself.

Multiple Bond Composition

- ▶ Single bond: **1** σ
- ▶ Double bond: **1** σ + **1** π
- ▶ Triple bond: **1** σ + **2** π

Key Distinction

The **first** bond is always σ ; additional bonds are π . π bonds are **weaker** — side-by-side overlap is less effective than end-to-end overlap.



Pi (π) bonds form from the side-by-side overlap of two *p* orbitals. The dots indicate the location of the nuclei.

Which Bond is Stronger: σ or π ?

🔑 A σ bond is **stronger** than a π bond because **head-on overlap** is more effective than **side-by-side overlap**.

σ Bond

- ▶ Formed by **head-on** overlap of orbitals.
- ▶ **Greater orbital overlap.**
- ▶ More electron density **between the nuclei.**
- ▶ **Stronger** bond.

π Bond

- ▶ Formed by **side-by-side** overlap of orbitals.
- ▶ **Less orbital overlap.**
- ▶ Electron density above and below the bond axis.
- ▶ **Weaker** bond.

Greater orbital overlap $\implies$ Stronger bond
 σ overlap $>$ π overlap $\therefore$ σ bonds are stronger than π bonds.

Check Your Understanding

For each molecule below, state the number of σ bonds and π bonds:

- (a) N_2 (triple bond)
- (b) CO_2 (two double bonds)
- (c) H_2O (two single bonds)
- (d) Can a molecule have a π bond *without* a σ bond between the same atoms?

Come to lecture to see the answer.

Hybrid Atomic Orbitals

Hybrid Atomic Orbitals



Orbitals as Vectors

Wavefunctions Form a *Vector Space*

An atomic orbital is a wavefunction $\psi(\mathbf{r})$. Since wavefunctions satisfy the superposition principle, if ψ_1 and ψ_2 are orbitals, then

$$a\psi_1 + b\psi_2$$

is also a **valid wavefunction** for any scalars a, b .

A **basis** is a set of vectors that

- ▶ are linearly independent (i.e. cannot be used to represent each other),
- ▶ span a vector space (i.e. can be used to represent any point in the vector space).

Every vector in that space can be written as a unique linear combination of the basis vectors.



Orbitals are simply vectors in an (infinite-dimensional) vector space of wavefunctions.



Example: A Four-Dimensional Subspace

The Valence Orbitals of Carbon

For covalent bonding in carbon, we focus on the four valence orbitals

$$\{2s, 2p_x, 2p_y, 2p_z\}.$$

These orbitals span a particular four-dimensional subspace we can call V

$$V = \text{span}\{2s, 2p_x, 2p_y, 2p_z\}.$$

Coordinates in This Basis

Every vector in V can be written **uniquely** as

$$\phi = a(2s) + b(2p_x) + c(2p_y) + d(2p_z).$$

The coefficients (a, b, c, d) are simply the coordinates of ϕ in this basis.

⚠ We are **not** describing all possible orbitals (e.g. $3s, 5d$); we are only describing those in the subspace relevant to carbon's valence bonding.

Hybridization as a Change of Basis

The orbitals $\{2s, 2p_x, 2p_y, 2p_z\}$ are **one basis** for the subspace V . Since a vector space admits many different bases, we are free to choose another orthonormal basis spanning the same subspace.

Hybrid Orbitals: Using CH_4 as an Example

This basis produces orbitals concentrated in different regions of space while preserving the occupied orbital subspace.

$$h_1 = \frac{1}{2}(2s + 2p_x + 2p_y + 2p_z),$$

$$h_2 = \frac{1}{2}(2s + 2p_x - 2p_y - 2p_z),$$

$$h_3 = \frac{1}{2}(2s - 2p_x + 2p_y - 2p_z),$$

$$h_4 = \frac{1}{2}(2s - 2p_x - 2p_y + 2p_z).$$

This corresponds to a new basis:
 $\{sp_1^3, sp_2^3, sp_3^3, sp_4^3\}.$

These four hybrid orbitals:

- ▶ are symmetry-equivalent (in CH_4 ; not NH_3 , H_2O , CH_3Cl , etc.)
- ▶ have identical shape
- ▶ point toward the four hydrogens

One localized representation describes each C–H bond as arising from overlap between one C sp^3 hybrid and one H $1s$ orbital.

Because methane has tetrahedral symmetry, the four localized C–H bonds are equivalent.

Hybridization is just a change of basis. The occupied valence orbital subspace is unchanged; only the orbital basis used to represent the wavefunction changes.

Why Use Hybridization as a Model?

The Problem with Pure Atomic Orbitals

Carbon's electron configuration is $1s^2 2s^2 2p^2$.

Pure p orbitals are at 90° to each other (p_x , p_y , p_z), but CH_4 bond angles are **109.5°** .

Hybridization explains this. Use it to predict and describe bonding more accurately.

Hybridization is a mathematical model showing an atom's independent bonding orbitals. Hybrid orbitals:

- ▶ Have **shapes distinct** from original s and p orbitals
- ▶ Are **equivalent in energy and shape** to each other
- ▶ Number equal to the orbitals combined
- ▶ Point toward bonding partners — **maximizing overlap**

Hybridization is not a physical process atoms perform. It is simply a **mathematical description** of bonding orbitals — we apply it only to atoms **once they are bonded**.

Hybridization and σ , π Bonding: Each Addresses Different Questions

Question	Determined By
Which orbitals are involved in bonding?	Hybridization (sp , sp^2 , sp^3 , ...)
How do those orbitals overlap?	σ vs. π bonding

Hybridization

- ▶ Mixes atomic orbitals to create equivalent hybrid orbitals.
- ▶ Determines **geometry** around an atom.
- ▶ Examples:
 - ▶ $sp \rightarrow$ linear
 - ▶ $sp^2 \rightarrow$ trigonal planar
 - ▶ $sp^3 \rightarrow$ tetrahedral

σ and π Bonds

- ▶ **σ bond:** head-on orbital overlap.
- ▶ **π bond:** side-by-side overlap of unhybridized p orbitals.

Bond	Composition
Single	1σ
Double	$1 \sigma + 1 \pi$
Triple	$1 \sigma + 2 \pi$

sp Hybridization: Linear Geometry

Formation of *sp* Hybrid Orbitals

Mixing **one** *s* + **one** *p* orbital produces **two** *sp* hybrids:

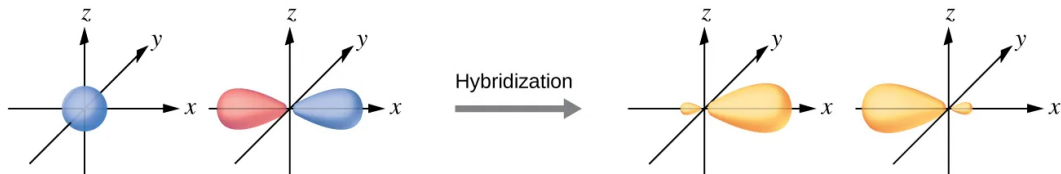
- ▶ Two lobes point in **opposite directions** (180° apart)
- ▶ Each hybrid points along the internuclear axis
- ▶ Two unhybridized *p* orbitals remain, perpendicular to the axis

Example: BeCl₂

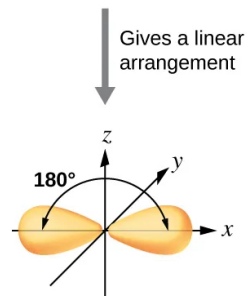
Be: $[He] 2s^2 \xrightarrow{\text{hybridize}} 2 \text{ } sp \text{ orbitals}$

Each *sp* orbital overlaps a Cl 3*p* orbital $\Rightarrow$ 2 σ bonds at 180°.

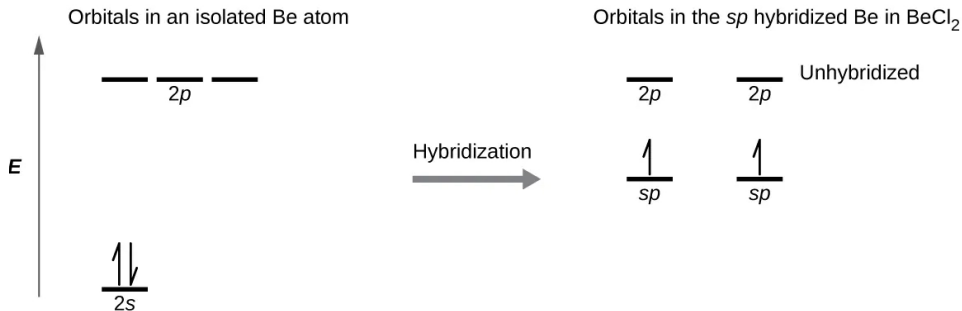
sp Hybridization: Linear Geometry



An *s* orbital (blue sphere) and a *p* orbital (shown with red and blue lobes representing opposite phases) hybridize to form two *sp* hybrid orbitals (yellow). Each hybrid orbital is oriented primarily in just one direction. Note that each *sp* orbital contains one lobe that is significantly larger than the other. The set of two *sp* orbitals are oriented at 180° , which is consistent with the geometry for two domains.



sp Hybridization: Linear Geometry



This orbital energy-level diagram shows the *sp* hybridized orbitals on Be in the linear BeCl_2 molecule. Each of the two *sp* hybrid orbitals holds one electron and is thus half filled and available for bonding via overlap with a Cl $3p$ orbital.

☠ Hybrid orbitals are mathematical combinations of atomic orbitals rather than orbitals with their own energies. Hybrid energy-level diagrams are only schematic.

sp^2 Hybridization: Trigonal Planar Geometry

Formation of sp^2 Hybrid Orbitals

Mixing **one** s + **two** p orbitals produces **three** sp^2 hybrids:

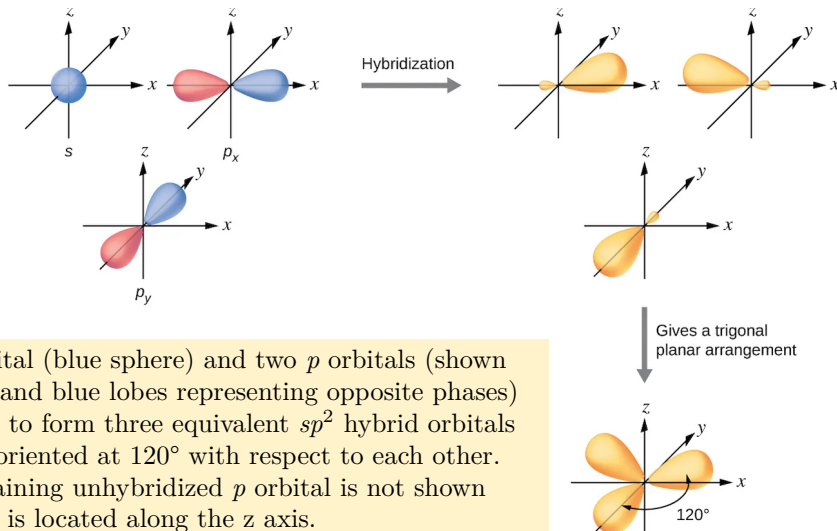
- ▶ Three lobes in a **trigonal planar** arrangement (120°)
- ▶ One unhybridized p orbital remains, perpendicular to the plane
- ▶ This p orbital is available for π bonding

Examples: BF_3 , C_2H_4

Each B in BF_3 : 3 sp^2 orbitals $\Rightarrow$ 3 σ bonds at 120° .

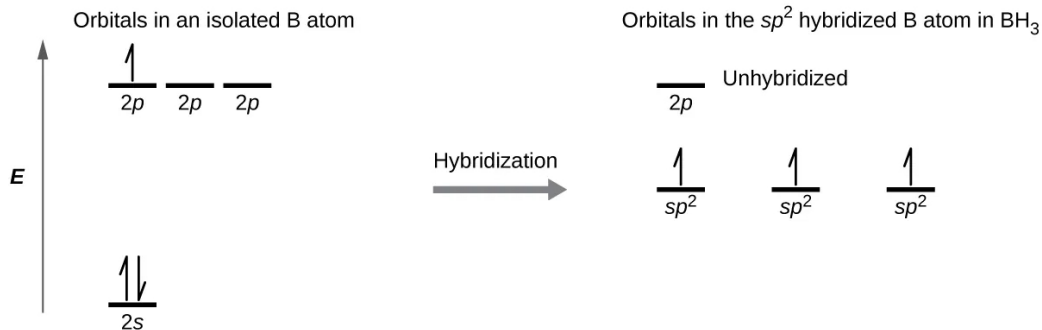
Each C in C_2H_4 : 3 sp^2 σ bonds + 1 unhybridized p for the π bond.

sp^2 Hybridization: Trigonal Planar Geometry



An s orbital (blue sphere) and two p orbitals (shown with red and blue lobes representing opposite phases) hybridize to form three equivalent sp^2 hybrid orbitals (yellow) oriented at 120° with respect to each other. The remaining unhybridized p orbital is not shown here, but is located along the z axis.

sp^2 Hybridization: Trigonal Planar Geometry



In an isolated B atom, there are one $2s$ and three $2p$ valence orbitals. When boron is in a molecule with three regions of electron density, three of the orbitals hybridize and create a set of three sp^2 orbitals and one unhybridized $2p$ orbital. The three half-filled hybrid orbitals each overlap with a $1s$ valence orbital from a hydrogen atom to form three σ bonds in BH_3 .

sp^3 Hybridization: Tetrahedral Geometry

Formation of sp^3 Hybrid Orbitals

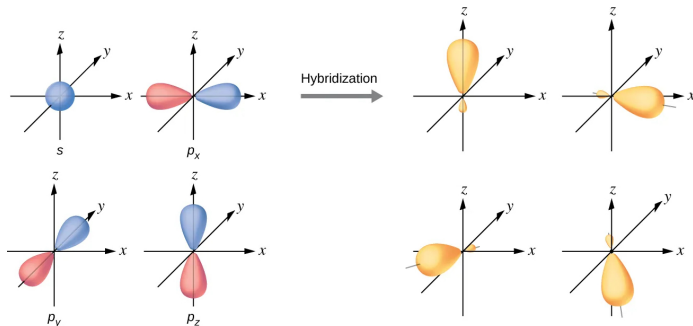
Mixing **one** s + **three** $p \Rightarrow$ **four** sp^3 hybrids:

- ▶ Four lobes point toward the corners of a **tetrahedron**
- ▶ Bond angles: **109.5°** ; no unhybridized p orbitals remain
- ▶ **Lone pairs** also occupy hybrid orbitals — they count as domains

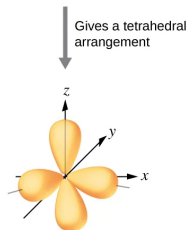
Examples

- ▶ CH_4 : 4 sp^3 hybrids $\Rightarrow$ 4 σ bonds at 109.5°
- ▶ NH_3 : 3 sp^3 bonds + 1 sp^3 lone pair $\Rightarrow$ pyramidal
- ▶ H_2O : 2 bonds + 2 lone pairs $\Rightarrow$ bent (104.5°)

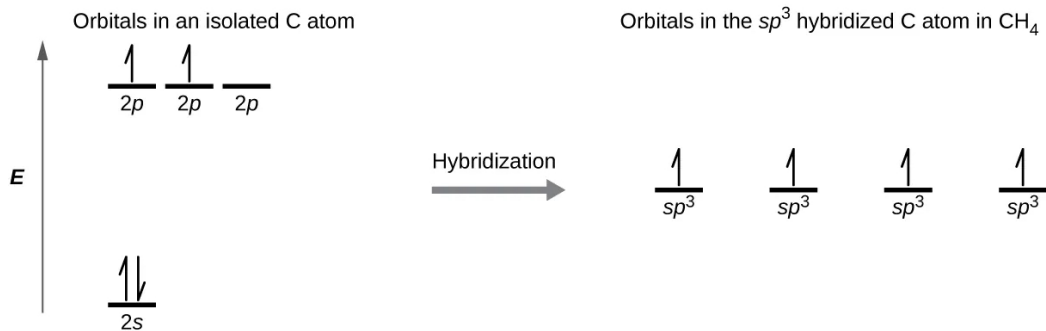
sp^3 Hybridization: Tetrahedral Geometry



An s orbital (blue sphere) and three p orbitals (shown with red and blue lobes representing opposite phases) hybridize to form four equivalent sp^3 hybridized orbitals (yellow) oriented at 109.5° with respect to each other.



sp^3 Hybridization: Tetrahedral Geometry



The four valence atomic orbitals from an isolated carbon atom all hybridize when the carbon bonds in a molecule like CH_4 with four regions of electron density.

This creates four equivalent sp^3 hybridized orbitals. Overlap of each of the hybrid orbitals with a hydrogen orbital creates a C–H σ bond.



Atom “In Itself” and “In Relation”: Same Atom, Different Lens

As an analogy, Hegel distinguishes between something considered *in itself* (*an sich*) and something understood *through its relations* (*für anderes*).

A carbon atom considered in isolation is naturally described by its atomic orbitals

$$\{2s, 2p_x, 2p_y, 2p_z\},$$

since we are considering carbon *by itself*.

But when carbon is considered *through its relations*, as part of a molecule such as CH_4 , we are no longer interested in the isolated atom, but in its role within the molecular whole. The same valence orbital subspace can then be represented using a different basis; one convenient choice is the set of sp^3 hybrid orbitals directed toward the four C–H bonds.

- ▶ Hybridization: **intratomic mixing** (of one atom's orbitals)
- ▶ Bonding: **interatomic mixing** (different atoms' orbitals into molecular orbitals).



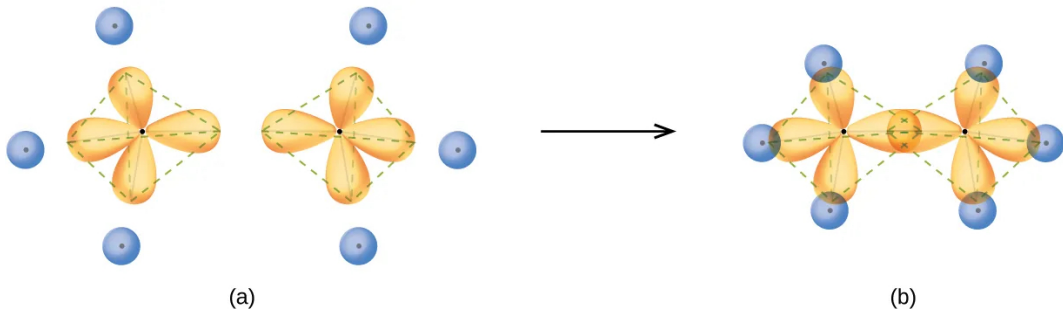
The carbon atom has not become a different atom.



What changes is the **most useful mathematical description**:

isolated atom $\implies$ atomic-orbital basis bonded atom $\implies$ one convenient hybrid basis.

sp^3 Hybridization: Tetrahedral Geometry — C_2H_6 Example



(a) In the ethane molecule, C_2H_6 , each carbon has four sp^3 orbitals.

(b) These four orbitals overlap to form seven σ bonds.

Check Your Understanding

Determine the hybridization of the **central atom** in each molecule:

- (a) SO_3 (trigonal planar, no lone pairs on S)
- (b) PF_3 (pyramidal shape)
- (c) XeF_2 (linear molecule, 3 lone pairs on Xe)

Come to lecture to see the answer.

sp^3d and sp^3d^2 Hybridization

sp^3d : Trigonal Bipyramidal (5 domains)

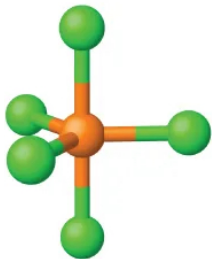
- ▶ Mix $s + 3p + d \Rightarrow$ **5 hybrids**
- ▶ 3 equatorial (120°) + 2 axial (90°)
- ▶ Example: PCl_5 , SF_4 , ClF_3 , XeF_2

sp^3d^2 : Octahedral (6 domains)

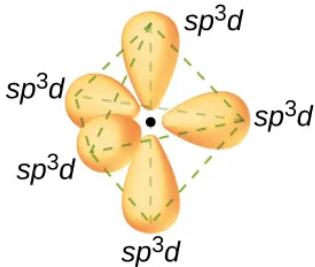
- ▶ Mix $s + 3p + 2d \Rightarrow$ **6 hybrids**
- ▶ All bonds at 90° — perfectly symmetric
- ▶ Example: SF_6 , PCl_6^- , XeF_4

Requires Period 3+ atoms. This is an *introductory* model; modern descriptions of hypervalent molecules often use **MO theory**.

sp^3d Hybridization: PCl_5 Example



(a)

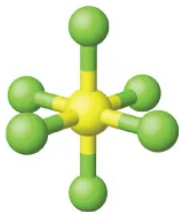


(b)

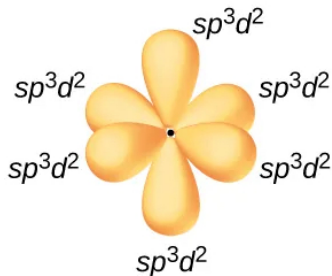
(a) The five regions of electron density around phosphorus in PCl_5 require five hybrid sp^3d orbitals.

(b) These orbitals combine to form a trigonal bipyramidal structure with each large lobe of the hybrid orbital pointing at a vertex. As before, there are also small lobes pointing in the opposite direction for each orbital (not shown for clarity).

sp^3d^2 Hybridization: SF_6 Example



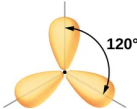
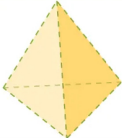
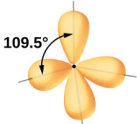
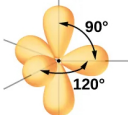
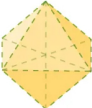
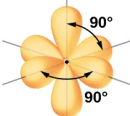
(a)



(b)

(a) Sulfur hexafluoride, SF_6 , has an *octahedral* structure that requires sp^3d^2 hybridization.

(b) The six sp^3d^2 orbitals form an octahedral structure around sulfur. Again, the minor lobe of each hybrid orbital is not shown for clarity.

Regions of Electron Density	Arrangement		Hybridization	
2		linear	sp	
3		trigonal planar	sp^2	
4		tetrahedral	sp^3	
5		trigonal bipyramidal	sp^3d	
6		octahedral	sp^3d^2	

How to Determine Hybridization



Strategy: Count Electron Domains on the Central Atom

1. Draw the Lewis structure.
2. Count **electron domains** (groups) on the central atom:
bonds (single, double, or triple each count as **one domain**) + lone pairs.
3. Match the number to the hybridization:

Domains	Hybridization	Geometry	Example
2	sp	linear	BeCl_2 , CO_2
3	sp^2	trigonal planar	BF_3 , C_2H_4
4	sp^3	tetrahedral	CH_4 , H_2O
5	sp^3d	trig. bipyramidal	PCl_5
6	sp^3d^2	octahedral	SF_6

Make sure you relate this to VSEPR theory!

Part IV

Multiple Bonds

Outline of Part IV: Multiple Bonds

Multiple Bonds

Outline of Part IV: Multiple Bonds

Multiple Bonds

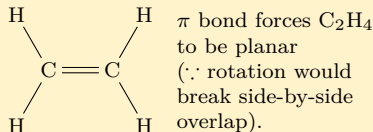
Studying Hybridization in Ethene (C₂H₄): A Double Bond

sp^2 Hybridization in Ethene

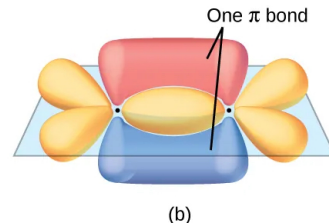
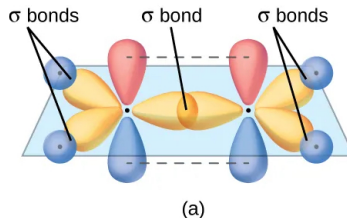
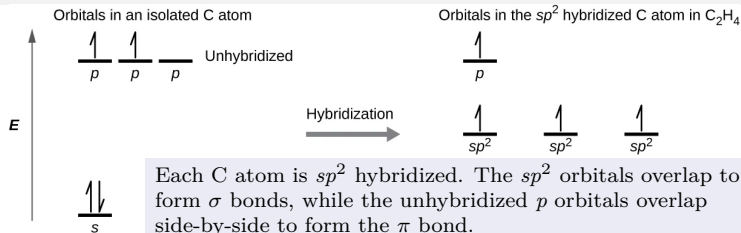
Each C in C₂H₄ is sp^2 hybridized:

- ▶ Three sp^2 orbitals form σ bonds: C–C and two C–H
- ▶ 1 unhybridized p on each C overlaps **side-by-side** $\Rightarrow$ π bond
- ▶ $\Rightarrow$ all atoms are in 1 plane

C₂H₄ double bond $\Rightarrow$ 1 σ + 1 π



single bond $\approx$ swivel joint
double bond $\approx$ locked hinge.

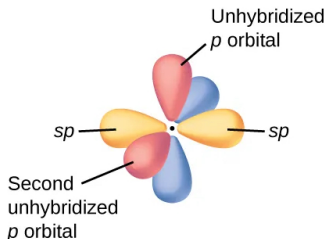


- (a) Five σ bonds form: one C–C bond from overlap of two sp^2 orbitals, and four C–H bonds from overlap of C sp^2 orbitals with H s orbitals.
- (b) The π bond forms by side-by-side overlap of the two unhybridized p orbitals, with electron density above and below the plane of the σ system.

Studying Hybridization in Acetylene (C_2H_2): A Triple Bond

sp Hybridization in C_2H_2

Each C is sp hybridized:

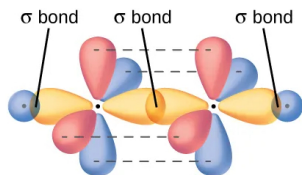


- ▶ Two sp : head-on overlap $\Rightarrow$ C–H and C–C σ bonds
- ▶ **Two** unhybridized p orbitals on each C
- ▶ These four p orbitals overlap side-by-side $\Rightarrow$ **two π bonds**

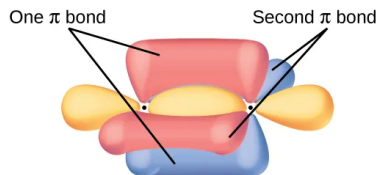
Result: *Linear* Molecule

We saw that ethene's unhybridized p_z orbital had lobes in the $+z$ and $-z$. Due to the spatial orientation of this orbital, the electron density is focused above and below the xy plane, which contains all the σ bonds.

For acetylene, this effect is even more pronounced: the p_y and p_z orbitals form a cylindrical electron-density cloud around the x axis, fixing the σ bonds in a *line*.



(a)



(b)

- (a) Two C–H σ bonds and one C–C bond consisting of one σ and two π bonds. The π bonds form by side overlap of four unhyb. p orbitals.
- (b) The two π bonds lie in perp. planes around the C–C σ bond.

Why Unhybridized p Orbitals Constrain Bond Geometry

💡 Imagine a water-filled balloon.

When you apply a constrictive force from the top and bottom, the water-filled balloon constricts in the direction of applied force (z axis) and expands leftward & rightward (x axis) and forward & backward (y axis).

When you apply a constrictive force from the top, bottom, front, and back, the water-filled balloon constricts in the direction of applied force (z axis and y axis) and expands leftward & rightward (x axis) only.

Since electrons exert force on other electrons, this analogy is useful in understanding how unhybridized p orbitals, necessarily perpendicular to the internuclear axis, force σ bonds either into a plane (in the case of a double bond) or into the line of the internuclear axis (in the case of a triple bond).

Restricted Rotation About Double Bonds

π Bonds Restrict Rotation

Rotation about a double bond would require **breaking the π bond**: the p orbitals must remain parallel to overlap. Bond energies show the cost:

Bond	Length (pm)	Energy (kJ/mol)
C–C (single)	150.6	347
C=C (double)	133.5	614
C $\equiv$ C (triple)	120.8	839

Geometric Isomers are a Consequence of this Restricted Rotation

The restricted rotation about C=C means **cis** and **trans** isomers of alkenes are distinct molecules — they cannot interconvert at room temperature.

Why this matters: the rigidity of C=C is why *unsaturated* fats (cis double bonds, kinked) behave differently from *saturated* fats (all single bonds, flexible).

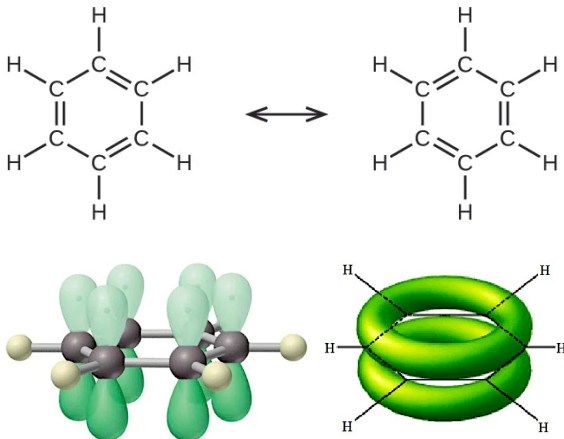
Benzene and Electron Delocalization: Resonance Explained

Benzene: sp^2 Carbon Ring

Each of the 6 carbon atoms in C_6H_6 is sp^2 hybridized:

- ▶ 3 sp^2 orbitals form σ bonds (C–C and C–H)
- ▶ 1 unhybridized p orbital on each C, all perpendicular to the ring
- ▶ All 6 p orbitals overlap to form a **delocalized π system**

Electron density is **spread equally** over the ring — the molecule does *not* switch back and forth between resonance structures; it is a single **hybrid** of them. Benzene is more **stable** than either Lewis structure predicts.



Hybridization and Resonance: A Key Rule

Assigning Hybridization in Resonance Structures

When determining hybridization, count σ **bonds**, **lone pairs**, and **unpaired electrons** only — the arrangement of π bonds does not matter.

Example: Ozone (O_3)

- ▶ O_3 has 2 resonance structures, each showing one $\text{O}=\text{O}$ and one $\text{O}-\text{O}$.
- ▶ Central O: 2 σ bonds + 1 lone pair = 3 domains $\Rightarrow sp^2$
- ▶ The π electrons are **delocalized** over both $\text{O}-\text{O}$ bonds.

Rule: hybridization of the central atom is the **same in all resonance forms** of a molecule.

Check Your Understanding

For **formaldehyde** ($\text{H}_2\text{C}=\text{O}$):

- (a) What is the hybridization of the carbon atom?
- (b) How many σ bonds and π bonds does the molecule have?
- (c) What is the $\text{H}-\text{C}-\text{H}$ bond angle?

Come to lecture to see the answer.

Part V

Molecular Orbital Theory

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Limitations of Valence Bond Theory

What VB Theory Cannot Explain

- ▶ VB localizes electrons **between pairs of atoms** — but many molecules have electrons **delocalized over the whole molecule**.
- ▶ VB cannot predict the **paramagnetic behavior of O_2** .

Molecular Orbital (MO) Theory

MO theory treats electrons as **delocalized** throughout the entire molecule, occupying orbitals that span all atomic nuclei.

These molecular orbitals arise from the linear combination of **atomic orbitals (AO)**.

VB focuses on *bonds* (where electrons pair between atoms).

MO focuses on *where electrons are allowed to exist* across the whole molecule.

Atomic Orbitals are Wavefunctions

An atomic orbital is a **wavefunction** (ψ)
(not a planetary orbit!)

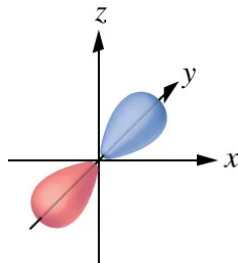
Like all **waves**, wavefunctions have:

- ▶ an **amplitude**
- ▶ a **phase**
 - ▶ often shown as + and –
 - ▶ these indicate the **sign of the wave**
 - ▶ they do NOT represent positive or negative charge

When two orbitals overlap, their wavefunctions combine according to

$$\psi_{\text{total}} = \psi_A + \psi_B$$

The resulting wave can be *larger* or *smaller* depending on the relative phase.



Blue and red (or + and –)
indicate opposite phases.

Interference happens when Waves (Orbitals) Overlap

Two overlapping wavefunctions interfere *constructively* or *destructively*.

Constructive interference

- ▶ waves are “**in phase**”
- ▶ amplitudes add
- ▶ larger overall wave amplitude

Destructive interference

- ▶ waves are “**out of phase**”
- ▶ amplitudes cancel
- ▶ wave amplitude decreases

Constructive Interference



+



Wave
amplitudes
add

Destructive Interference



+



Wave amplitudes
cancel to give
zero amplitude

Why Interference Matters for Chemistry

Electrons occupy **wavefunctions**, so interference actually changes the likelihood of electrons being found in a certain region (electron density).

This redistribution of electron density (a consequence of interference) is what causes bonding and antibonding interactions.

Constructive interference

- ▶ increases wave amplitude
- ▶ increases “electron density” (probability of electron existing) between nuclei
- ▶ lowers the potential energy
- ▶ promotes bonding

Destructive interference

- ▶ decreases wave amplitude
- ▶ decreases electron density between nuclei
- ▶ raises the potential energy
- ▶ can produce a **node**, a region where an electron cannot exist

Nodes

Whenever two atomic orbitals overlap,

$$\psi_{\text{MO}} = \psi_A + \psi_B$$

There are always two possible combinations:

- ▶ constructive interference → **bonding orbital** (σ , π)
- ▶ destructive interference → **antibonding orbital** (σ^* , π^*)

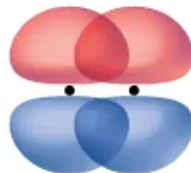
A **node** is a region where

$$\psi_{\text{MO}} = 0$$

because waves cancel by destructive interference.

At a node,

- ▶ the wavefunction changes sign
- ▶ the probability of finding an electron is zero
- ▶ electron density is absent



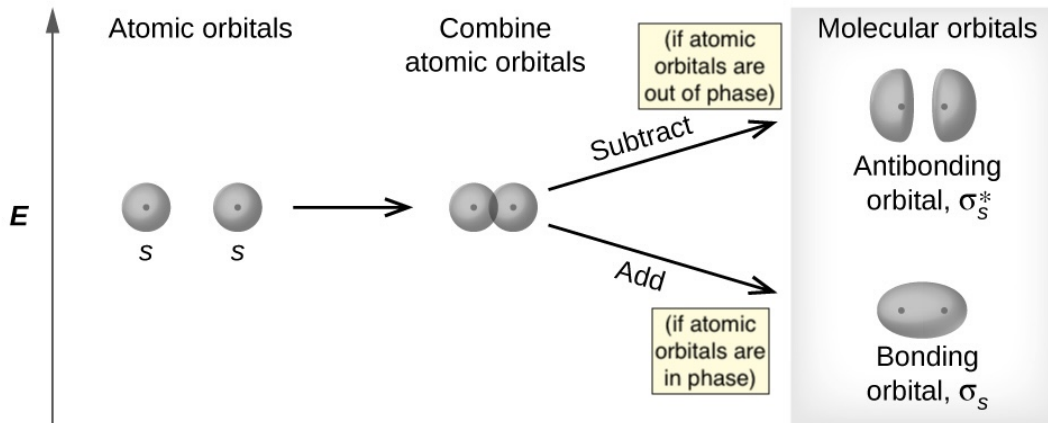
The resulting wave (ψ_{MO}) **changes phase** as it passes through the node.

Nodes occur in **antibonding orbitals**.

Interference of Two s Orbitals

Constructive $\rightarrow$ Bonding Orbital, σ_s

Destructive $\rightarrow$ Antibonding Orbital, σ_s^*

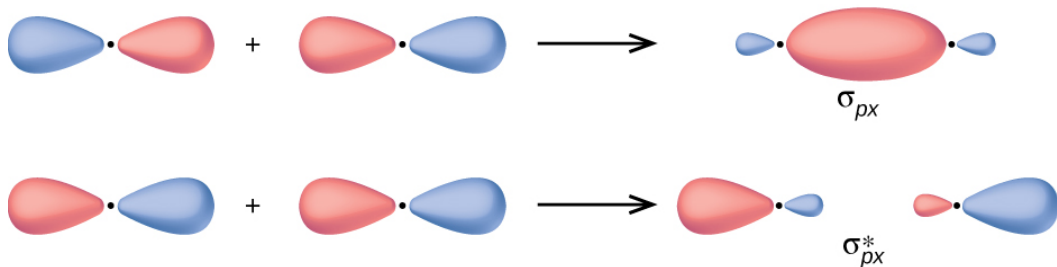


 For the *added* atomic orbitals (bonding orbital, σ_s), should the region between the nuclei be darker, indicating a greater likelihood of finding an electron there?

Interference of Two p Orbitals Forming a σ bond

Constructive $\rightarrow$ Bonding Orbital, σ_{px}

Destructive $\rightarrow$ Antibonding Orbital, σ_{px}^*



Blue and red are used to indicate opposite phases.

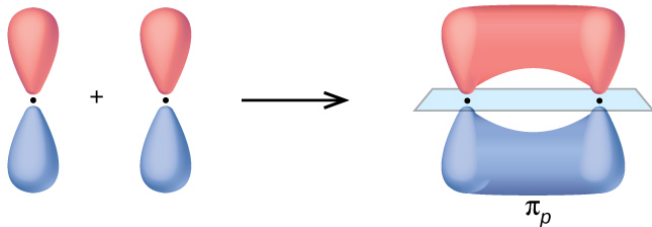
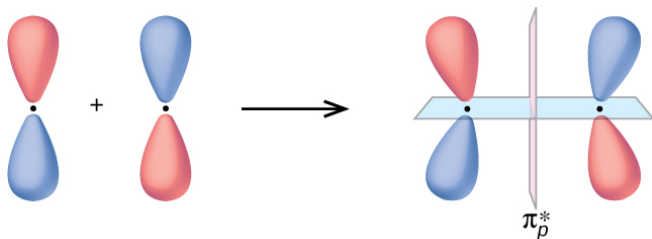
We can consider the blue orbitals to be + wavefunctions ($\psi > 0$) and red orbitals to be - wavefunctions ($\psi < 0$).

- ▶ Electron density is greatest between the nuclei for the bonding orbital σ_{px} (indicating a bond).
- ▶ Electron density is zero between the nuclei for the antibonding orbital σ_{px}^* (no chance of an e^-).

Interference of Two p Orbitals Forming a π bond

Constructive $\rightarrow$ Bonding Orbital, σ_s

Destructive $\rightarrow$ Antibonding Orbital, σ_s^*



Blue and red are used to indicate opposite phases.

We can consider the blue lobes of the orbitals to be $+$ ($\psi > 0$) and the red lobes of the orbitals to be $-$ ($\psi < 0$).

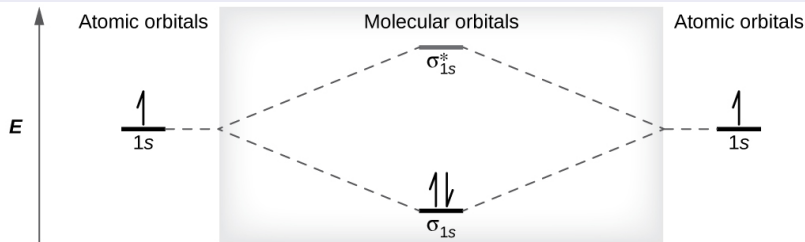
- ▶ Electron density is greatest between the nuclei for the bonding orbital π_p (indicating a bond).
- ▶ Electron density is zero between the nuclei for the antibonding orbital π_p^* (no chance of an e^-).

Bonding and Antibonding Orbitals

How MOs Form from AOs

When two atomic orbitals combine, they produce **two molecular orbitals**:

- ▶ **Bonding MO** (σ , π):
constructive interference $\Rightarrow e^-$ density **between** nuclei; **lower** PE than original AOs.
- ▶ **Antibonding MO** (σ^* , π^*):
destructive interference $\Rightarrow$ a **node between** nuclei; **higher** potential energy.



Electrons in **bonding MOs** *stabilize* the molecule.

Electrons in **antibonding MOs** *destabilize* it.

Bond Order

Definition: Bond Order

Bond order is a measure of how many net bonding pairs exist between two atoms:

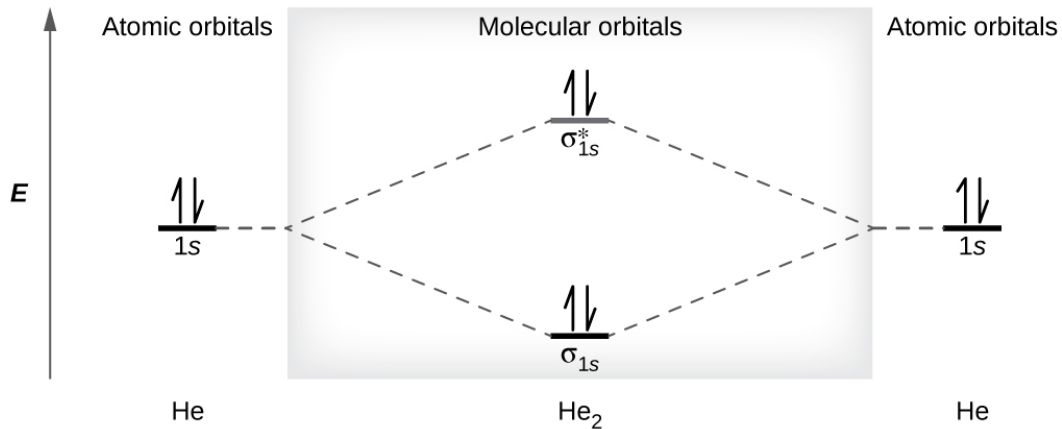
$$\text{Bond Order} = \frac{(\text{bonding electrons}) - (\text{antibonding electrons})}{2}$$

Interpreting Bond Order

- ▶ Bond order = 0: **no stable bond** (molecule does not exist)
- ▶ Bond order = 1, 2, 3: single, double, triple bond
- ▶ **Higher bond order $\Rightarrow$ shorter bond $\Rightarrow$ more energy to break it** (these move together — don't memorize separately)

$$\text{H}_2: (\sigma_{1s})^2 \Rightarrow \text{Bond Order} = \frac{2 - 0}{2} = \boxed{1} \quad (\text{stable single bond})$$

He₂: Why Helium Does Not Dimerize



He₂: Why Helium Does Not Dimerize

MO Configuration of He₂

He atom: $1s^2$ — so He₂ would have 4 electrons total.

MO filling: $(\sigma_{1s})^2 (\sigma_{1s}^*)^2$

$$\text{Bond Order} = \frac{2 - 2}{2} = \boxed{0}$$

Conclusion

$\text{Bond order} = 0 \Rightarrow \text{He}_2 \text{ is not a stable molecule.}$

The bonding and antibonding orbitals **cancel each other out** exactly.

This explains why noble gases exist as **monatomic** species — no net bonding is possible.

Check Your Understanding

The **hydrogen molecule ion** H_2^+ has only **one electron**.

- (a) Write its MO electron configuration.
- (b) Calculate the bond order.
- (c) Is H_2^+ stable? Is it more or less stable than H_2 ?

Come to lecture to see the answer.

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

MO Theory for Second-Period Diatomics

Bridge: the number of orbitals increases (now $2s$ and three $2p$ on each atom), but the *rules stay identical* — combine each pair of AOs into a bonding and an antibonding MO, then fill them by Aufbau/Hund/Pauli.

Orbitals Involved

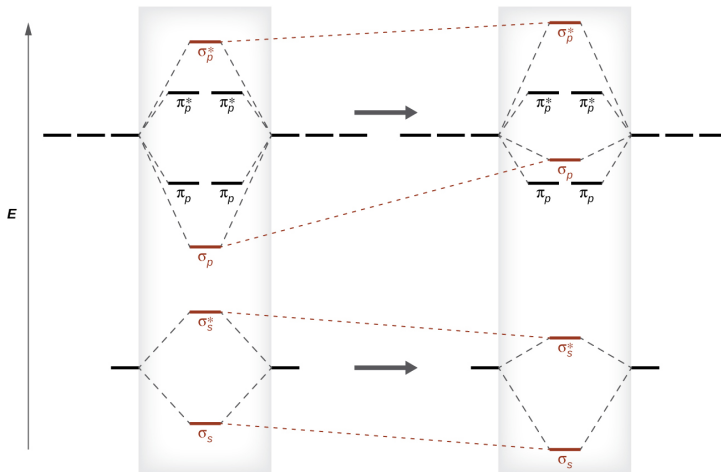
For Period-2 homonuclear diatomics (Li_2 through Ne_2), both $2s$ and $2p$ atomic orbitals contribute MOs:

- ▶ $2s$ AOs $\Rightarrow \sigma_{2s}$ and σ_{2s}^*
- ▶ $2p$ AOs (end-to-end) $\Rightarrow \sigma_{2p}$ and σ_{2p}^*
- ▶ $2p$ AOs (side-by-side) $\Rightarrow$ two π_{2p} and two π_{2p}^*

$s-p$ Mixing

For **lighter** Period-2 elements (Li_2 through N_2), the σ_{2s} and σ_{2p} energies are close enough to mix, pushing the π_{2p} orbitals **lower in energy** than σ_{2p} . For **O_2 , F_2 , Ne_2** : no $s-p$ mixing; σ_{2p} is **lower** than π_{2p} .

$s-p$ Mixing (occurs when the s and p orbitals have similar energies)



The energy difference between $2s$ and $2p$ orbitals in O, F, and Ne is greater than that in Li, Be, B, C, and N. $\therefore$ $s-p$ mixing is not seen in O, F, and Ne. It *is* seen in the other period-2 elements.

The σ_s wavefunction mathematically combines with the σ_p wavefunction, with the result that the σ_s orbital becomes more stable, and the σ_p orbital becomes less stable.

Similarly, the antibonding orbitals also undergo $s-p$ mixing, with the σ_s^* becoming more stable and the σ_p^* becoming less stable.

Without mixing, the MO pattern occurs as expected, with the σ_p orbital lower in energy than the π_p orbitals.

When $s-p$ mixing occurs, the orbitals shift as shown, with the σ_p orbital higher in energy than the π_p orbitals.

MO Energy Diagram: O₂

MO Filling Rules

Fill MOs by **Aufbau** (lowest energy first), **Hund's rule** (degenerate orbitals fill singly first), and **Pauli exclusion**.

N₂ (10 valence electrons)

$$(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$$

$$\text{Bond Order} = \frac{8 - 2}{2} = \boxed{3} \text{ (triple bond)}$$

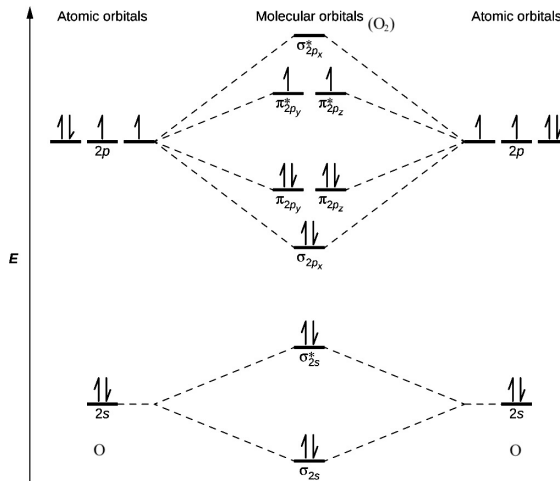
All electrons paired $\Rightarrow$ **diamagnetic**

O₂ (12 valence electrons)

$$(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p})^2(\pi_{2p})^4(\pi_{2p}^*)^2$$

$$\text{Bond Order} = \frac{8 - 4}{2} = \boxed{2}$$

2 unpaired electrons $\Rightarrow$ **paramagnetic**



O₂ Paramagnetism: The Mystery Solved

Why O₂ Is Paramagnetic

The two highest-energy electrons in O₂ occupy the **degenerate** π_{2p}^* **orbitals** one each (Hund's rule) $\Rightarrow$ **two unpaired electrons**.

Unpaired electrons have a magnetic moment $\Rightarrow$ the molecule is attracted to magnetic fields.

Lewis Structure vs. MO Theory

- ▶ Lewis structure: O=O (all paired) — **predicts diamagnetic**
- ▶ MO theory: 2 unpaired e^- in π^* — **predicts paramagnetic**

This is a fundamental victory of MO theory over simpler bonding models. It correctly predicts a measurable, real-world property that Lewis theory predicts incorrectly.

Summary: Properties of Second-Period Diatomics

Molecule	Valence e^-	Bond Order	Unpaired e^-	Magnetism
Li ₂	2	1	0	diamagnetic
Be ₂	4	0	0	(does not exist)
B ₂	6	1	2	paramagnetic
C ₂	8	2	0	diamagnetic
N ₂	10	3	0	diamagnetic
O ₂	12	2	2	paramagnetic
F ₂	14	1	0	diamagnetic
Ne ₂	16	0	0	(does not exist)

Pattern: bond strength increases toward N₂ because *bonding* electrons are being added; past N₂ the new electrons go into *antibonding* π^* orbitals, so bond order drops back. N₂ is the strongest bond in chemistry.

Check Your Understanding

Using MO theory, determine for F_2 :

- (a) The bond order.
- (b) Whether F_2 is paramagnetic or diamagnetic.
- (c) How many σ and π bonds does F_2 have?

Come to lecture to see the answer.

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Three Theories: A Comparison

Chemists choose a model based on the question being asked.

Feature	Lewis / VSEPR	VB Theory	MO Theory
e^- location	shared pairs	localized between atom pairs	delocalized over molecule
Geometry	predicted (VSEPR)	via hybridization	via electron count in MOs
Paramagnetism	cannot predict	cannot predict	predicts correctly
Resonance	multiple structures	multiple structures	single wave function
Best for	connectivity	geometry, hybridization	mag. / elec. properties

All three are **approximations**. None is completely accurate: each is simply a tool optimized for different questions.

Key Terms and Summary

VB Theory & Hybridization

- ▶ **Valence bond theory:** bonds via orbital overlap
- ▶ **σ bond:** end-to-end overlap; all single bonds
- ▶ **π bond:** side-by-side overlap; 2nd/3rd bonds
- ▶ **Hybridization:** mixing AOs to form hybrid orbitals
- ▶ sp , sp^2 , sp^3 , sp^3d , sp^3d^2 : hybrid types

MO Theory

- ▶ **MO theory:** electrons delocalized over molecule
- ▶ **Bonding MO:** stabilizes molecule (lower potential energy)
- ▶ **Antibonding MO (*):** destabilizes (higher potential energy)
- ▶ **Bond order:** $\frac{\text{bonding} - \text{antibonding}}{2}$
- ▶ **Paramagnetic:** unpaired electrons
- ▶ **Diamagnetic:** all electrons paired

Final Check Your Understanding

Without drawing a full MO diagram, predict whether each species is **stable**, **paramagnetic**, and give its **bond order**:

(a) O_2^{2-} (peroxide ion — add 2 electrons to O_2)

(b) N_2^+ (remove 1 electron from N_2)

Come to lecture to see the answer.